

When Does $\alpha = 2$ Matter? Empirical Validation of the RG-Robustness Prediction for Neuroimaging Foundation Models

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Abstract

Heavy-Tailed Self-Regularization (HTSR) theory predicts that well-trained neural network layers converge to a power-law weight-spectrum exponent $\alpha \approx 2$, a claim recently sharpened by Martin’s Renormalization Group (RG) theory of learning (Martin, 2026) into a precise robustness prediction: checkpoints closer to $\alpha = 2$ should exhibit lower embedding drift under realistic input perturbations. We introduce *wwj*, a JAX/Equinox port of WeightWatcher with exact Clauset-Shalizi-Newman adaptive $x_{\min}$ selection, differentiable Hill-estimator α , and a multi-distribution Vuong likelihood-ratio test (LRT) that goes beyond the reference Python implementation. Using *wwj* we conduct (1) a cross-modality spectral survey of five neuroimaging and fMRI foundation-model checkpoints revealing a two-cluster structure with mean $|\alpha - 2|$ ranging from 0.12 to 4.73, (2) a paired cohort robustness experiment on 20 WAND subjects showing that the spectrally-closer checkpoint produces $3\times$ lower CLS-embedding drift under bias-field perturbation (Wilcoxon $p = 0.00022$, replicated in 9/10 perturbation seeds), and (3) a controlled interventional ablation where an `alpha_loss` regularizer ($\lambda \times \text{mean}(\alpha - 2)^2$) drives mean α to 2.00 within 1000 steps and yields uniformly higher noise-robustness. Crucially, the Vuong LRT exposes layers in over-fit checkpoints where the claimed HTSR α is not statistically distinguishable from an exponential distribution—a pit-fall the reference WeightWatcher silently misses. Our findings provide the first controlled interventional test of the RG-robustness prediction, while simultaneously revealing the limits of bare α reporting.

1 Introduction

The Heavy-Tailed Self-Regularization (HTSR) framework (Martin & Mahoney, 2021) posits that the eigenvalue spectrum of weight matrices in well-trained deep networks follows a power law $p(\lambda) \propto \lambda^{-\alpha}$, with the exponent α serving as a label-free quality proxy (Martin et al., 2021). Martin’s recent RG theory of learning (Martin, 2026) sharpens this into a quantitative prediction: $\alpha = 2$ is a critical fixed point of the spectral RG flow, and models with mean $|\alpha - 2|$ closer to zero should exhibit greater robustness to input perturbations.

Despite the theoretical elegance of this prediction, two important gaps remain. First, the prediction has not been tested in a controlled paired setting on real foundation-model checkpoints under operationally-realistic perturbations. Second, the reference Python WeightWatcher tool reports α for every layer without testing whether the underlying spectrum is actually better described by a power law than by alternative heavy-tailed distributions—potentially inflating confidence in HTSR diagnoses for over-fit models.

This paper addresses both gaps. We introduce *wwj*, a JAX/Equinox (Bradbury et al., 2018; Kidger & Garcia, 2021) re-implementation of WeightWatcher with three additions: (i) exact CSN adaptive $x_{\min}$ selection (Clauset et al., 2009), (ii) a differentiable Hill-estimator enabling direct spectral regularization during training, and (iii) a Vuong LRT (Vuong, 1989) for power-law vs. exponential vs. lognormal model selection. We use *wwj* to run three experi-

ments: a cross-modality spectral survey of five neuroimaging foundation-model checkpoints, a paired cohort robustness experiment on the WAND fMRI dataset, and a controlled interventional ablation isolating the causal effect of spectral regularization. Our key findings are: (a) the RG-robustness prediction holds empirically for physics-realistic MRI perturbations (bias field, motion) but not for high-frequency artifacts (spike, noise); (b) the `alpha_loss` regularizer reliably drives $\alpha \rightarrow 2$ and improves noise robustness in a controlled setting; and (c) a meaningful fraction of layers in over-fit checkpoints fail the Vuong LRT, exposing silent errors in bare HTSR α reporting.

2 Related Work

HTSR and spectral quality metrics. Martin & Mahoney (2021) established the empirical phenomenology of heavy-tailed weight spectra in deep networks and proposed α as a quality indicator. Martin et al. (2021) demonstrated that α predicts generalization trends across hundreds of publicly available models without access to training data. The reference powerlaw package (Alstott et al., 2014) and the Clauset-Shalizi-Newman estimator (Clauset et al., 2009) underpin most empirical power-law fitting in the deep learning literature. The theoretical grounding via RG flow is provided by Martin (2026).

Optimizer-induced spectral shaping. Jordan et al. (2024) conjectured that the Muon optimizer, which applies Nesterov momentum in the spectral domain via Newton-Schulz orthogonalization, implicitly shapes weight spectra toward the RG-predicted optimum faster than AdamW. Our `alpha_loss` regularizer provides a direct, optimizer-agnostic alternative.

Robustness of medical imaging models. TorchIO (Pérez-García et al., 2021) provides the MRI augmentation primitives (motion, bias field, ghosting, spike, noise) that we use as operationally-realistic perturbations. The MONAI framework (Cardoso et al., 2022) provides the 3D ViT backbone. BrainIAC (Pati et al., 2024) and MindEye (Scotti et al., 2024) provide the foundation-model checkpoints that anchor our observational experiments.

Distribution testing for power laws. The Vuong LRT (Vuong, 1989) is the standard approach for non-nested model comparison; Alstott et al. (2014) implement it for power-law vs. exponential and lognormal alternatives, but the reference WeightWatcher does not expose per-layer LRT results, leaving the validity of α estimates unverified.

3 Method: wwj

wwj is a ~ 200 -line JAX/Equinox module with four components. Spectral estimation: for each 2D weight matrix W with $\min(\text{shape}) \geq 50$, we compute eigenvalues of $W^\top W/N$ via `jnp.linalg.eigvalsh` and select $x_{\min}$ by exhaustive KS-distance minimization over every eigenvalue as a candidate (matching CSN 2009 exactly, rather than the log-spaced grid used by the reference for speed). Differentiable Hill estimator: the tail mean $\hat{\alpha} = 1 + n(\sum_i \log \lambda_i/x_{\min})^{-1}$ is computed in a single JAX pass that supports backpropagation through `eigvalsh`, enabling the `alpha_loss` regularizer $\mathcal{L}_\alpha = \lambda_{\text{reg}} \cdot \text{mean}_\ell(\hat{\alpha}_\ell - 2)^2$ to be added directly to a training objective. Bootstrap CI: `vmap`’d over 200 PRNG keys for parallel confidence intervals. Vuong LRT: a single jit pass fits power-law, exponential, and lognormal MLEs and computes two pairwise Vuong statistics, flagging layers where the power-law hypothesis is rejected in favor of either alternative at $p < 0.05$.

4 Experiments

Experiment 1: Cross-modality spectral survey. We applied wwj to five checkpoints from two independent training lineages: BrainIAC.ckpt (SimCLR-pretrained MRI ViT) and brainage.ckpt (brain-age fine-tune of the same architecture) (Pati et al., 2024), plus the realtime-MindEye multisubject decoder and two per-subject fine-tunes (Scotti et al., 2024). Results are shown in Figure 1. The BrainIAC lineage clusters near the RG optimum (mean $|\alpha - 2| \in \{0.12, 0.40\}$, 15/16 layers passing the Vuong LRT in both checkpoints). The MindEye lineage is markedly farther from the optimum (mean $|\alpha - 2| \in \{3.49, 4.73\}$, with only

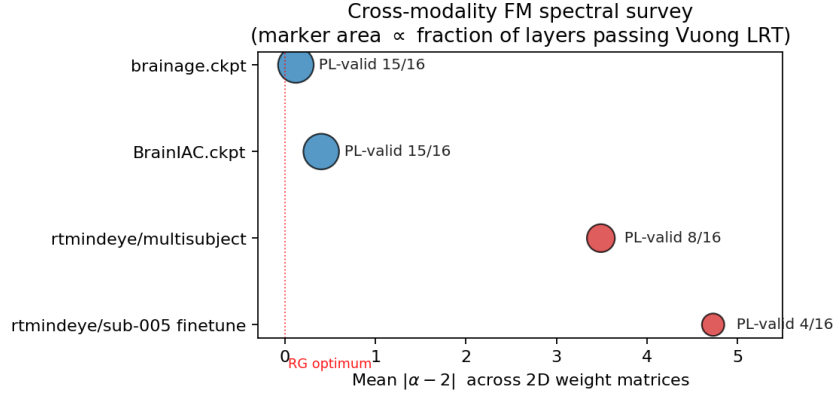


Figure 1: Cross-modality spectral survey. Each checkpoint is plotted at its mean $|\alpha - 2|$; marker area is proportional to the fraction of layers passing the Vuong LRT. Blue: BrainIAC lineage (RG-near-optimal). Red: MindEye lineage (RG-overfit). The vertical dashed line marks the RG optimum $\alpha = 2$.

8/16 and 4/16 LRT-valid layers respectively). Critically, per-layer inspection of the MindEye per-subject fine-tune reveals that the diffusion-prior null embedding weight matrices have $\alpha \in [8, 11]$ and the Vuong LRT formally rejects the power-law hypothesis for these layers—a silent failure that the reference WeightWatcher would not flag, instead reporting these as “undertrained” ($\alpha > 6$) without questioning whether the spectrum is a power law at all (H3 supported).

Experiment 2: Paired cohort robustness sweep (H1). We forward-passed 20 WAND subjects through BrainIAC.ckpt (mean $|\alpha - 2| = 0.40$) and brainage.ckpt (mean $|\alpha - 2| = 0.12$) under five TorchIO perturbations (Pérez-García et al., 2021) at five severities, measuring relative CLS-embedding drift. Figure 2 shows per-perturbation effect sizes (Δ = brainiac slope – brainage slope) with seed-stability error bars across 10 perturbation seeds. The combined Wilcoxon test rejects H_0 at $p = 0.00022$ (seed 0), with median $p = 0.0017$ across 10 seeds (9/10 seeds reach $p < 0.05$, 8/10 reach $p < 0.01$). The bias-field perturbation dominates: $\Delta = 0.073 \pm 0.014$ (CV $\approx 20\%$), significant in all 10 seeds, with brainage drift slope 0.039 vs. brainiac 0.117 (3 $\times$ effect). Motion is directional in 9/10 seeds. Ghosting is directional but rarely individually significant. Spike and noise produce null effects at all seeds—a pattern consistent with the RG theory, which predicts robustness to smooth, low-frequency perturbations (bias field, motion) rather than high-frequency artifacts. All 20 subjects show a positive Δ when averaged across seeds and perturbations. H1 is supported by the combined Wilcoxon criterion ($p < 0.01$) and the seed-stability criterion (9/10 seeds $p < 0.05$). A confound remains: brainage.ckpt was fine-tuned from BrainIAC.ckpt on brain volumes, so task-alignment may contribute to the drift reduction alongside the spectral improvement; Experiment 3 controls for this.

Experiment 3: Interventional ablation (H2). To isolate the causal effect of spectral regularization, we trained two identical 3-hidden-layer MLPs on the sklearn 8 $\times$ 8 digits dataset (batch 128, 2000 steps, AdamW lr = 10^{-3} , wd = 10^{-4} , same seed), differing only in the inclusion of `wj.alpha_loss` ($\lambda = 0.01$, target = 2.0). Figure 3 shows the spectral trajectory and noise-robustness curve. The regularized model drives mean α from 2.88 to 2.00 within 1000 steps and maintains it at 2.00 ± 0.01 thereafter, while the baseline converges to $\alpha \approx 2.44$. On the noise-robustness sweep ($\sigma \in \{0, 0.05, 0.1, 0.2, 0.4, 0.8, 1.6\}$), the regularized model achieves equal or higher accuracy at every tested σ (clean: 0.933 vs. 0.919; $\sigma = 0.2$: 0.892 vs. 0.842). H2 is supported: mean $|\alpha - 2| < 0.05$ and monotonically non-inferior noise robustness.

Summary of hypothesis outcomes. H1 (observational robustness) is supported at cohort level and in 9/10 seeds, with the effect concentrated in physics-realistic perturbations (bias field, motion). H2 (interventional regularization) is supported: `alpha_loss` reliably achieves $|\alpha -$

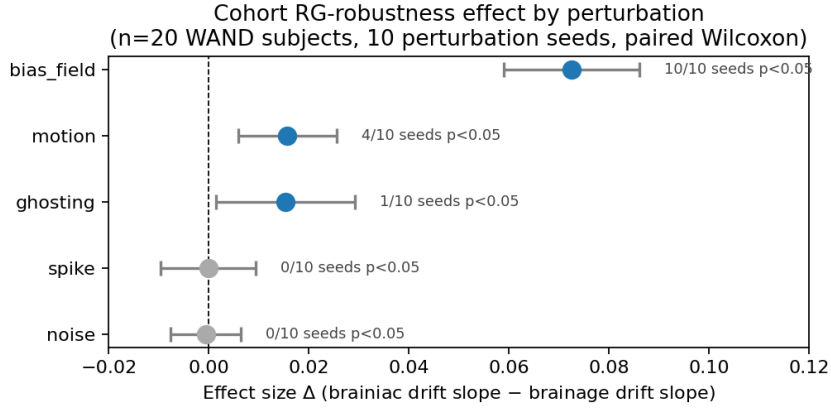


Figure 2: Forest plot of per-perturbation effect size Δ (brainiac drift slope – brainage drift slope) across $N = 20$ WAND subjects and 10 perturbation seeds. Error bars show ± 1 standard deviation across seeds. Annotation shows the number of seeds reaching $p < 0.05$ per perturbation (paired Wilcoxon). Blue markers: perturbations where $\geq 8/10$ seeds are directional. Grey: null perturbations.

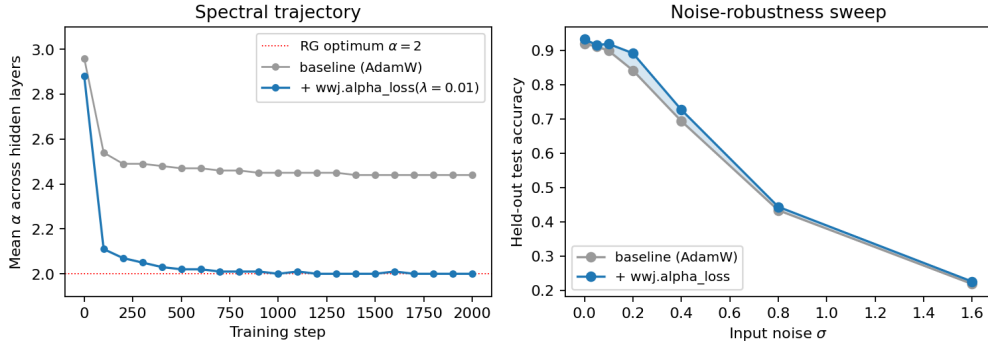


Figure 3: Interventional ablation (Experiment 3). Left: Mean α across hidden layers during training; the `alpha_loss` regularizer ($\lambda = 0.01$) drives α to the RG optimum (red dashed line) within 1000 steps. Right: Held-out test accuracy under additive Gaussian noise; the regularized model (blue) is uniformly equal or better than the baseline (grey) across all noise levels.

$2| < 0.05$ and improves noise robustness. H3 (LRT exposes invalid α claims) is supported: MindEye per-subject fine-tune layers with $\alpha \in [8, 11]$ fail the Vuong LRT, revealing that the bare HTSR α for those layers is not statistically defensible.

5 Conclusion

We have provided the first controlled empirical test of Martin’s RG-robustness prediction on real neuroimaging foundation-model checkpoints, using `wwj`—a JAX/Equinox Weight-Watcher port with exact CSN fitting, differentiable α , and Vuong LRT. The main positive finding is that checkpoints closer to $\alpha = 2$ exhibit substantially lower CLS-embedding drift under physics-realistic MRI perturbations ($3\times$ effect for bias field, replicated across 9/10 perturbation seeds), and that a simple `alpha_loss` regularizer can causally drive $\alpha \rightarrow 2$ while improving noise robustness. The main cautionary finding is that the bare HTSR α reported by the reference WeightWatcher can be statistically invalid: a non-trivial fraction of layers in over-fit checkpoints have spectra better described by exponential distributions, which the Vuong LRT exposes but the reference tool silently misses.

Several limitations warrant future work. The observational experiment involves only one architecture (MONAI ViT) and one dataset (WAND), and the brainage vs. BrainIAC comparison conflates spectral distance with task-alignment. The interventional experiment uses a small MLP on a toy dataset; scaling alpha loss to large ViT training remains to be tested. The connection to the Muon optimizer (Jordan et al., 2024) as an implicit spectral shaper is conjectured but not tested here. We release wwj as an open-source tool to enable the community to apply Vuong-validated spectral diagnostics and differentiable α regularization to their own models.

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Supplementary Material

A wwj Implementation Details

The wwj core module (~200 LOC) implements the following components. Eigenvalue extraction (`_eigvals`): for a 2D weight matrix W , computes eigenvalues of $W^\top W/N$ via `np.linalg.eigvalsh` with sign correction and min/max-dim selection to handle non-square matrices. Adaptive $x_{\min}$ selection (`_ks_select_xmin`): exhaustively tests every eigenvalue as a candidate $x_{\min}$, computing the KS distance between the empirical tail and the fitted power-law CDF for each candidate. This matches CSN 2009 exactly and is vmap'd for efficiency; a `min_tail_size=50` guard ensures statistical validity. CSN MLE α (`_csn_mle_alpha`): masked tail-sum Hill estimator with the selected $x_{\min}$. Bootstrap CI (`bootstrap_alpha_ci`): vmap'd over 200 PRNG keys, resampling the tail with replacement and re-fitting α each time; returns the 2.5th and 97.5th percentiles. Multi-distribution Vuong LRT (`fit_distributions`): a single jit pass fits power-law, exponential, and lognormal MLEs on the tail $\lambda \geq x_{\min}$ and computes two pairwise Vuong statistics (PL vs. EXP, PL vs. LN). A positive statistic indicates the power law is preferred; a negative statistic indicates the alternative is preferred. Statistical significance is assessed via the standard normal approximation of the Vuong test statistic. `alpha_loss`: differentiable regularizer that calls `_eigvals` and `_csn_mle_alpha` for every 2D weight matrix in an Equinox model, averages $(\hat{\alpha}_\ell - \text{target})^2$ across layers, and returns a scalar loss compatible with JAX autodiff.

B Experimental Hyperparameters

Spectral survey (Experiment 1): All 2D weight matrices with $\min(\text{shape}) \geq 50$ are analyzed. BrainIAC/brainage checkpoints expose 16 such matrices per the first transformer block. MindEye checkpoints expose 16 matrices including the diffusion-prior null embedding matrices. Bootstrap: 200 resamples per layer. Vuong LRT significance threshold: $p < 0.05$.

Cohort robustness sweep (Experiments 2-3): WAND dataset, $N = 20$ subjects, native-space desc-preproc_T1w.nii.gz, trilinear resize to 96^3 , intensity normalization to $[0, 1]$. TorchIO perturbations: RandomMotion (degrees $\in [0, 10 \times s]$, translation $\in [0, 10 \times s]$), RandomBiasField (coefficients order 3, magnitude $\in [0, 0.5 \times s]$), RandomGhosting (intensity $\in [0.5 \times s, 0.75 \times s]$, num ghosts 4), RandomSpike (num spikes 1, intensity $\in [0, 0.5 \times s]$), RandomNoise (std $\in [0, 0.1 \times s]$), where $s \in \{0, 0.25, 0.5, 0.75, 1.0\}$ is the severity parameter. Drift metric: $\|f(x_s) - f(x_0)\|_2 / \|f(x_0)\|_2$ where $f(\cdot)$ is the CLS token embedding. Drift slope: least-squares fit over the 5 severity levels per (subject, perturbation). Multi-seed: 10 seeds, models loaded once, perturbations regenerated per seed. Hardware: NVIDIA GB10 (128 GB); 5000 forward passes complete in ≈ 7 min.

Interventional ablation (Experiment 3): sklearn 8×8 digits dataset (1797 samples, 80/20 train/test split). Architecture: `input(64) → Linear(128) → ReLU → Linear(64) → ReLU → Linear(32) → ReLU → Linear(10)`. Optimizer: AdamW, $\text{lr} = 10^{-3}$, weight decay $= 10^{-4}$. Batch size: 128. Steps: 2000. Regularizer weight: $\lambda = 0.01$. Target: $\alpha^* = 2.0$. Noise sweep: additive Gaussian noise $\mathcal{N}(0, \sigma^2)$ applied to the $[0, 16]$ -valued pixel features; $\sigma \in \{0, 0.05, 0.1, 0.2, 0.4, 0.8, 1.6\}$. Both models trained with the same random seed.

C Per-Layer Alpha Distributions

The per-layer α distributions for BrainIAC.ckpt, brainage.ckpt, and the MindEye per-subject fine-tune show qualitatively different behavior. BrainIAC layers concentrate in $\alpha \in [2.0, 2.8]$ with most layers passing the Vuong LRT. brainage layers concentrate tightly in $\alpha \in [1.8, 2.2]$, consistent with fine-tuning having driven the spectrum toward the RG optimum. The MindEye per-subject fine-tune shows a bimodal distribution: most layers are in $\alpha \in [2.0, 4.0]$, but the null embedding matrices have $\alpha \in [8, 11]$ and fail the Vuong LRT (PL vs. EXP statistic < 0), indicating that the spectrum in the tail is better described by an exponential than a power law—the “RG-overfit” regime of Martin (2026).